CP–6209

Sub. Code
11

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015.

Aircraft Maintenance Science

ENGLISH

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Give an account of Mrs. Packletide's Tiger.

2. Explain the deed of bravery.

3. Who is Ozymandias?

4. Describe Ulysses.

5. What is waterloo?

6. What happened on Mother's Day?

7. Define Comprehension.

8. What are communication skills?

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Narrate the importance of water as the elixir of life.

Or

(b) Describe the journey to Niagara.

10. (a) How does Goldsmith depict the Village School Master?

Or

(b) What is the central idea in "The Solitary Reaper"?

11. (a) Explain the patriotic spirit revealed in "The Rising of the Moon".

Or

(b) Discuss the plot of "The Proposal".

12. (a) Develop the hints into an essay :

Two friends - went to forest - suddenly a bear - came near - one boy climbed the tree quickly - escape from the bear - another boy - helpless - lay down on the ground silently - bear came close - watched - went away - the boy on the tree came down - asked him - what did bear say - in his ears - boy replied - dont believe friends who leave you when in trouble.

Or

(b) Write a letter to your sister informing her for your success in the Bank exam.

13. (a) Discuss the need for effective communication skills.

Or

(b) Prepare a welcome speech for your college day.

CP–6210

Sub. Code
12

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aircraft Maintenance Science

MATHEMATICS

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Write down the middle term of $\left(x-\frac{2}{x}\right)^{12}$.

2. Show that $(a-b)^n = a^n\left[1-n\left(\frac{b}{a-b}\right)+\frac{n(n+1)}{2!}\left(\frac{b}{a-b}\right)^2-\cdots\right]$.

3. Find $\lim_{x\to 0}\frac{x-\sin x}{x^3}$.

4. If $\frac{\sin\theta}{\theta}=\frac{5045}{5046}$, prove that the angle θ is $1^0 58^1$, nearly.

5. If $y=\frac{x+1}{x+2}$, find d^2y/dx^2.

6. Evaluate $\int\frac{x^2\tan^{-1}x}{1+x^2}dx$.

7. Solve $p^3-7p-6=0$.

8. Solve $y=3x+\log p$.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Find the coefficient of x^n in the expansion of $\frac{x-2}{(x+2)(x-1)^2}$ in ascending powers of x.

Or

(b) Find the sum to infinity of the series

$$\frac{5}{3.6}+\frac{5.7}{3.6.9}+\frac{5.7.9}{3.6.9.12}+\cdots.$$

10. (a) Prove that

$$\cos^4\theta\sin^3\theta=\frac{-1}{2^6}\left[\sin 7\theta+\sin 5\theta-3\sin 3\theta-3\sin\theta\right].$$

Or

(b) Find, $\lim_{x\to 0}\left(\frac{1}{x^2}-\frac{1}{\sin^2 x}\right)$.

11. (a) If $y=e^x\tan^{-1}x$, then prove that $(1-x^2)y_2-(1-x+x^2)y_1+(1-x)^2y=0$.

Or

(b) (i) Find n^{th} derivative of $\frac{x^2-4}{(x+1)(x+4)}$. (7)

(ii) Find the nth derivative of $\log\left(\frac{x+a}{x-a}\right)$. (5)

12. (a) (i) Evaluate $\int \left(x + \frac{1}{x}\right)^2 dx$. (4)

(ii) Evaluate the integral of the type $\int x\left(x^2+1\right)^{7/2} dx$. (4)

(iii) Evaluate $\int 5x\sqrt{1-2x^2}\, dx$. (4)

Or

(b) (i) Evaluate $\int \frac{\sin(\log x)}{x} dx$. (6)

(ii) Evaluate $\int \frac{2x+1}{(x-3)^2} dx$. (6)

13. (a) Solve $p^2 - 2py = 3y^2$, where $p = {}^{dy}/_{dx}$.

Or

(b) Solve $x p^2 + 2px - y = 0$, where $p = {}^{dy}/_{dx}$.

CP–6211

Sub. Code
13

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aircraft Maintenance Science

BASIC ELECTRICITY

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. What is diamagnetism? Explain.
2. Explain the term induced emf.
3. Why voltage regulators are necessary in rectifier circuit?
4. Briefly explain the working of a voltmeter.
5. What is an inductor? Give its units.
6. Explain alternating current using suitable graph.
7. Explain the peak value of an AC.
8. Describe Q-factor of a series LCR circuit.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Give the theory of diamagnetism.

Or

(b) (i) Discuss the various methods of producing voltage.

(ii) State and explain the laws of electromagnetic induction.

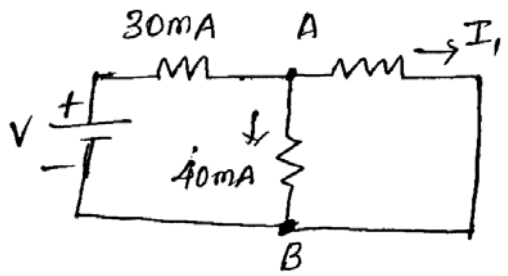
10. (a) (i) Describe the construction and uses of switches.

(ii) With a neat circuit diagram, explain the construction and working of a halfwave rectifier.

Or

(b) (i) State and explain Kirchoff's current and voltage law.

(ii) A series circuit of a gadget is as shown in figure below.



Find the value of the current I_1.

11. (a) Explain the construction, working and uses of a multimeter with necessary circuit diagram.

Or

(b) Give the explanation and describe the principle, construction and working of a AC generator.

12. (a) Discuss in detail about AC through pure ohmic resistance only.

Or

(b) Explain the term phase difference. Derive an expressing for the current through the circuit containing R and L in series.

13. (a) Give the theory of parallel resonance LCR circuit. Explain why it is called as the rejector circuit.

Or

(b) Obtain an expression for the current in an a.c circuit containing R and C in series.

CP–6212

Sub. Code
21

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aircraft Maintenance Science

PRINCIPLES OF ELECTRONICS AND ELECTRONIC CIRCUITS

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Define colour coding method and its uses.

2. Write short note on inductance.

3. Define forward voltage.

4. Define knee voltage.

5. Write short notes on FET.

6. List out the various types of amplifier.

7. Define distortion.

8. What do you meant by oscillators? Give few application of it.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Define resistor and explain various types of resistor.

Or

(b) Briefly explain the capacitor and its types.

10. (a) Describe the construction and operation of NPN transistors.

Or

(b) Explain the common emitter configuration of transistor with circuit diagram.

11. (a) Describe the construction and working of MOSFET.

Or

(b) Briefly explain the construction and working of UJT with necessary diagram.

12. (a) With a neat diagram explain the construction and operation of complementary pair amplifier.

Or

(b) Briefly explain the operation of class C amplifier also define power relation, load power and power dissipation.

13. (a) With neat circuit diagram explain the operation of Hartley oscillator.

Or

(b) Explain the operation of Colpitt's oscillator and also give few application of it.

CP–6213

Sub. Code
22

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aircraft Maintenance Science

APPLIED PHYSICS

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Write a short note on Newton rings.
2. Explain resolving power.
3. Discuss about Galiliean transformation.
4. What is meant by NDT and write down their advantages?
5. Mention the applications of laser in Engineering and medical field.
6. Define Total internal reflection in optical fibre.
7. Differentiate between crystalline and noncrystalline solids with suitable examples.
8. Briefly explain the super conductivity.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) What is thin film? How the fringes are produced by a wedge shaped thin film?

Or

(b) Discuss the Fraunhofer diffraction pattern due to a single slit.

10. (a) Describe the construction of a nicol prism and show how it can be used as a polarizer or as on analyser.

Or

(b) Derive Lorentz transformation equations for the space and time coordinates.

11. (a) Express the equation about sabine's formula.

Or

(b) Explain the principle, construction and working of magnetostriction oscillator method.

12. (a) Briefly discuss about the Holography.

Or

(b) Mention the types of optical fibre.

13. (a) What are the crystal systems in Bravass lattices?

Or

(b) Mention the types of crystal defects in detail.

CP–6214

Sub. Code
23

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aircraft Maintenance science

STRENGTH OF MATERIALS AND APPLIED MECHANICS

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Define rigid body, Scalar quantities.

2. Define moment of a force, vector quantities.

3. Define elastic limit, vector quantities.

4. Mention various types of loads.

5. Define Factor of safety.

6. Explain the two types of belt drives with sketches.

7. Explain composition of forces.

8. Define stress, strain and Elasticity.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Write short notes on Fundamental units and derived units.

Or

(b) Define Force and Explain various systems of forces.

10. (a) Explain stress – strain relationship of materials.

Or

(b) Explain stresses in nuts and bolts.

11. (a) Draw the shear force and Bending moment diagrams for the over hanging beam carrying a uniformly distributed load of W Kn/metre over the entire length as shown. Also locate the point of controflexture.

Or

(b) Draw the shear force and Bending moment diagram for the cantilever beam of Length 'L' Carrying a concentrated load 'W' of the free end.

12. (a) Explain design requirements of shafts and uses.

Or

(b) Explain rivetting and different types of rivets.

13. (a) Explain different types of fasteners used in aircraft construction.

Or

(b) Explain various Gear drives with neat diagrams and applications.

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CP–6215

Sub. Code
24

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aircraft Maintenance Science

FUNDAMENTALS OF THERMODYNAMICS

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Define the term thermal engineering.
2. What is meant by closed system?
3. Define PMM of second kind.
4. State Avogadro's law.
5. Define Reynolds number.
6. What is compressibility factor?
7. Define latent heat of vaporization.
8. What are the various types of gas power cycle?

Part B (5 × 12 = 60)

Answer **all** questions

9. (a) Explain and derive Isothermal process.

Or

(b) Briefly explain about thermodynamic equilibrium.

10. (a) Air in a closed vessel of fixed volume of 0.15 m^3, exerts, pressure of 12 bar at 250°C, if the vessel is cooled. So that the pressure falls to 3.5 bar, determine the final temperature, heat transfer and change of entropy.

Or

(b) Derive Clausius -Clapeyron equation.

11. (a) Derive and explain steady flow process.

Or

(b) One kg of air is compressed polytropically $(n = 1.3)$ from 1 bar and 27°C to 3 bar. Find

(i) work transfer

(ii) Heat transfer

(iii) change in interval energy.

12. (a) Briefly explain about radiation of heat exchangers.

Or

(b) Explain the conduction of heat exchanger.

13. (a) Explain the construction and working of dual cycle.

Or

(b) Explain with neat sketches of reversed heat engine.

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CP–6216

Sub. Code
31

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aircraft Maintenance Science

MECHANICS OF FLIGHT AND AIRCRAFT PERFORMANCE

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Define Atmosphere and its contents.
2. Write the significance of Adiabatic Process and atmosphere lapse rates.
3. Define induced drag and its contribution of total drag of the aircraft.
4. Write the significance of L/D Ratio.
5. Define the term "Load Factor".
6. Define dynamic Stability.
7. What is Vertical Pull-Down?
8. What is Phugoid motion?

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) List out the Physical Properties of Atmosphere.

Or

(b) Define the Newton's IIIrd law of motion and its application.

10. (a) List out and explain the factors contributing lift generation.

Or

(b) Explain the Aerofoil Terminology.

11. (a) What is Bernoulli's Equation and describe about the application?

Or

(b) Write short notes on :

(i) Effect of Aspect Ratio

(ii) Wing Flaps.

12. (a) Explain about the load factor contribution in airplane design.

Or

(b) Explain the forces on the airplane in flight.

13. (a) Describe about the take-off performance.

Or

(b) Write short notes on :

(i) Dutch Roll

(ii) Spiral Divergence related to lateral dynamic stability.

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CP–6217

Sub. Code
32

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aircraft Maintenance Science

FLUID MECHANICS

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Explain specific volume and specific gravity.

2. Sketch a mercury U tube manometer arrangement and explain its working.

3. Deduce Bernoulli's equation of motion from Euler's equation of motion.

4. How are turbines classified?

5. Give the principle of centrifugal pump. Mention its advantages.

6. What is the difference between a fluid coupling and fluid torque convertor?

7. Write short notes on direct acting hydraulic lift.

8. Write short notes on gear wheel pump.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Explain in detail about any three properties of fluids.

Or

(b) (i) Explain the term kinematic viscosity.

(ii) With a help of a neat diagram, explain the working of 'U' tube manometer and differential manometer.

10. (a) Derive Euler's equation using Newton's law. Show that Bernoulli's equation can be obtained from Euler's equation. How is Bernoulli's equation related to conservation of energy?

Or

(b) Define the following and give example for each.

(i) Laminer flow

(ii) Turbulent flow

(iii) Steady flow and

(iv) Uniform flow.

11. (a) Explain in detail the power transmission through pipes.

Or

(b) Explain water hammers and discuss any two related problems in detail.

12. (a) Define the reciprocating pump. Explain the working of a reciprocating pump with sketches. What are its applications?

Or

(b) Define the terms :

(i) Specific speed of a turbine

(ii) unit speed

(iii) unit power and

(iv) unit rate of flow of a turbine

Derive the expression for specific speed and unit speed.

13. (a) Explain with neat sketch the working of different hydraulic lifts. Mention its advantages.

Or

(b) Explain the principle and working of hydraulic crane with neat sketch.

CP–6218

Sub. Code
33

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aircraft Maintenance Science

AIRCRAFT MATERIALS AND NON DESTRUCTIVE TESTS

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. What is elasticity and its applicability to aircraft structure?
2. Describe brass , its composition and its uses.
3. Describe different rivets used in aircraft.
4. Distinguish exographs and gamma graphs.
5. How the wood can be bended?
6. Define normalizing.
7. What do you meant by 'mutu metal'?
8. Explain the internal structure of steel.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Define :

   Elasticity, ductility ,normalizing, Annealing, Yield strength yield point.

   Or

   (b) Explain the process involved in rockwell hardness test.

10. (a) Explain the properties and uses of alloys of aluminium.

Or

(b) Discuss SAE numbering system.

11. (a) Describe the annealing process and its types.

Or

(b) Explain the heat treatment process involved in hardening of steel.

12. (a) Define rivets its types and riveting procedure.

Or

(b) Name some defects in woods and how they can be controlled.

13. (a) Explain about dye penetrate inspection.

Or

(b) Explain boroscope inspection.

CP–6219

Sub. Code
34

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aircraft Maintenance Science

AIRFRAME STRUCTURE AND SYSTEMS

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Write the significance of fail safe design.
2. State the various forces which will be acting on the structure during flying and explain.
3. What is the purpose of torque link on landing gear assembly?
4. Name the primary and secondary flying controls of an aircraft.
5. What is understood by 'fly–by–wire' system?
6. Define ballast and its types.
7. What is the function of seals and how they are classified?
8. Write short note on fuel crossfeed system.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Explain the following terms:
   - (i) Fuselage station
   - (ii) Water line
   - (iii) Buttock line
   - (iv) Aileron station.

Or

(b) Explain the retraction systems of the landing gear.

10. (a) What are the various types of flaps used in aircraft? Explain.

Or

(b) With neat sketch, explain the operation of ailerons.

11. (a) Why the aircraft levelling is done? Explain various methods of aircraft levelling.

Or

(b) Explain the theory of weight and balance. What are the preparations to be carried out before weighing an aircraft?

12. (a) Describe the three principle types of hydraulic fluid used in aircraft? Explain its characteristics and properties.

Or

(b) State the advantages and disadvantages of pneumatic system over hydraulic system.

13. (a) Explain types of fuel tanks used in aircraft.

Or

(b) Describe light aircraft gravity feed fuel system with neat diagram.

CP–6220

Sub. Code
41

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aircraft Maintenance Science

PISTON ENGINE AND PROPELLER

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Differentiate the terms value lead and valve lag.

2. Explain the functions of crankcase.

3. Write a note on single piece fixed pitch metal propellar.

4. Why magneto Ignition is superior than Battery Ignition system?

5. What do you understand by heat range of spark plug?

6. Write a note on engine driven fual pump. (EDP).

7. Define

   (a) Propeller blade angle

   (b) Propeller blade station.

8. Write the advantages air cooling system.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Describe four stroke fire event petrol engine cycle with p.v. diagram.

Or

(b) Define :

(i) power

(ii) horse power

(iii) indicated torse power

(iv) frictional horse power

(v) staft horse power.

10. (a) Name the value operating mechanism parts and explain in detail about inline engine value operating mechanism.

Or

(b) Describe the construction and operation of turbo charger mechanism.

11. (a) Explain in detail about the construction of liquid cooling system of piston engine.

Or

(b) Name the various types of oil system. Explain in detail about dry sump lubrication system.

12. (a) Explain in detail about various types of propellers.

Or

(b) Describe the function of constant speed propeller.

13. (a) Explain the classification detail of magneto.

Or

(b) Write the construction detail of spark plug.

CP–6221

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42

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aircraft Maintenance Science

TURBINE ENGINE

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. List the main parts of a jet engine.
2. State Newton's laws of motion.
3. State the advantages and disadvantages of annular type combustion chamber.
4. Describe the function of fuel cooled oil cooler.
5. Describe the function of spur gear type fuel pump.
6. What is meant by 'by sump' oil system?
7. What are the principle application of the turbo shaft engine?
8. What is called power turbine?

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Describe briefly the events of the Brayton cycle with volume Vs Pressure and Temperature Vs Pressure diagram.

Or

(b) Describe the thrust equation of Gas Turbine engine.

10. (a) Describe the function of the turbine nozzle diaphragm (Turbine inlet guide Vanes) and method of nozzle vane cooling.

Or

(b) Describe the function and types of thrust reversers commonly being used on gas turbine engine.

11. (a) Describe the oil analysis methods and oil sample analysis report.

Or

(b) What are the characteristics of Jet fuel? Describe the various types of turbine fuel.

12. (a) Describe about the air starter system.

Or

(b) Describe the ignition system for a turbine engine.

13. (a) Describe the four categories of gas turbine engine.

Or

(b) What does an EPR measure indicates and torque meter indicates?

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CP–6222

Sub. Code
43

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aircraft Maintenance Science

INSTRUMENTS AND COMPASS

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. What are the four principle elements of an instrument?

2. What are the methods information is displayed in an instrument?

3. What are the effects on human body due to loss of oxygen in high attitude?

4. What do you understand about pitot-static system of an aircraft?

5. Describe Gimbal Loch Anil Gimbal Error in a displacement Gyroscope.

6. What are the Error's in a Direct Reading magnetic compass?

7. Describe types of thermo couple and its usage.

8. What is the purpose of erection mechanism in gyroscope. State the pneumatically operated artificial horizon. Erection system.

Part B (5 × 12 = 60)

Answer **all** questions

9. (a) What is the purpose of colour range markings on the dials of certain instrument?

Or

(b) Draw a neat diagram of pitot-static probes and list the parts of it.

10. (a) What are the code settings of an altimeter and its indications?

Or

(b) State the basic principle construction and operation of an Altimeter.

11. (a) Define the following :

(i) Gyro

(ii) Input AXIS and out put AXIS of Gyro

(iii) Gimbal rings of Gyro

(iv) Gyro drift.

(v) Standard atmospheric pressure

(vi) Kew Baro meter.

Or

(b) What do you understand about Aircraft-Magnetism and its effects on compasses?

12. (a) State the basic principle construction and operation of Tachometer and Generator.

Or

(b) Describe the basic principle, construction and operation of a CHT gauge.

13. (a) With a neat diagram state the construction and operation of continuous flow type oxygen system.

Or

(b) What do you understand by

(i) Detector unit

(ii) Anoxia

(iii) Compass rose

(iv) Deviation and variation

(v) Portable oxygen set

(vi) Fast Erection switch.

CP–6223

Sub. Code
44

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aircraft Maintenance Science

AIRCRAFT ELECTRICAL SYSTEM

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. What is called free electrons?

2. What is called electromotive force and its units?

3. What is electrolyte and its importance?

4. What is known as residual magnetism?

5. What is significance of reverse current cut-out relay?

6. What is electrical load analysis and its importance?

7. What is called bonding jumper?

8. List out the significance of exterior lighting circuits.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Explain the practical applications of Ohm's law.

Or

(b) Write short notes on units of electricity.

10. (a) Describe with neat sketch of lead-acid battery construction.

Or

(b) List out the procedures of nickel-cadmium battery maintenance.

11. (a) List out the characteristics of DC generators.

Or

(b) Describe the static inverters.

12. (a) Explain the control of the power distribution systems.

Or

(b) Write short notes on (i) wire and cable clamping (ii) routing of electric wire bundles.

13. (a) Write short notes on Anti-collision lights.

Or

(b) Write short notes on Proximity sensor.

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CP–6224

Sub. Code
51

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aircraft Maintenance Science

FUNDAMENTALS OF COMPUTER AND OFFICE AUTOMATION

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Write a note on the types of software.

2. Write the input units you know and state their purpose.

3. Write down the features of MS-Word document.

4. List out the most commonly used File menu options in MS Word.

5. Write the steps to align text in chart titles and text boxes.

6. Write the steps to number the slides in a PowerPoint presentation.

7. Write the steps to print the slides in a PowerPoint presentation.

8. Write the parts of Access window in MS Access.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Discuss on the characteristics of software and types of software.

Or

(b) Explain logic gates and its types with example.

10. (a) Discuss on the steps to create a word document in MS Word with example.

Or

(b) Explain the elements in standard and formatting toolbar of MS Word.

11. (a) Describe the procedure for the following with example :

(i) Entering numbers

(ii) Entering and aligning text and

(iii) Formula.

Or

(b) Discuss on the standard toolbar items in MS Excel.

12. (a) Explain the following :

(i) Drawing picture and format commands in MS PowerPoint and

(ii) Steps to create a new presentation and the saving with auto save option.

Or

(b) Write steps to create a PowerPoint presentation to advertise your own company with various animation effects.

13. (a) Write the steps to create a new table in student database in MS Access with example.

Or

(b) Explain the steps to create forms and reports in MS Access with example.

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CP–6225

Sub. Code
52

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aircraft Maintenance Science

AVIATION LEGISLATION — I

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. List out the duties of Nodal officer.

2. What are the precautions to be observed while positioning fuelling vehicle near the aircraft for fueling operation?

3. What do you mean by "Aerodrome" and "Airstrip"?

4. How the registration marks are affixed on airplane wings?

5. What are the registration markings that are not commonly allotted?

6. Write short notes on Aviation Fuels.

7. What do you know about "Concession" Related to aircraft maintenance?

8. Discuss in brief about flight manual.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Discuss in detail about Aircraft Log Books.

Or

(b) Elaborately explain about quality control manual.

10. (a) Explain in detail about civil aviation requirements.

Or

(b) Write all points related to aircraft flight manual.

11. (a) Explain in detail about certificate of registration.

Or

(b) Explain in detail about certificate of Airworthiness.

12. (a) List out the precautions to be observed while fuelling the Aircraft.

Or

(b) Discuss in detail about categories of Aircraft Registration in India.

13. (a) Explain in detail about storage of Aviation Fuel.

Or

(b) Discuss in detail about quality check of Fuel ; Starting from source of supply to receiving point (Aircraft Fuel Tank).

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aircraft Maintenance Science

AVIATION MAINTENANCE PRACTICES

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Describe the concept of progressive inspection.

2. Define major maintenance.

3. Write the classifications of landing gear.

4. Why it is necessary to bleed the brake system?

5. What is the significance of torque links?

6. Write short note on engine ignition harness.

7. Write the operating principles of propellers.

8. What is intergranular corrosion? How it occur?

Part B (5 × 12 = 60)

Answer **all** the questions.

9. (a) Explain 100 hrs and the Annual Inspections.

Or

(b) Describe the special checks on aircraft after a heavy landing.

10. (a) List out the occasions for landing gear retraction check. Explain the procedure and checks carried out during landing gear retraction check.

Or

(b) Explain how the fixed landing gear of an aircraft is maintained.

11. (a) Explain the steps for inspection of engine controls.

Or

(b) Explain the testing and storage procedures for ignition system components.

12. (a) Explain about the periodic maintenance, repair and installation of wooden propeller.

Or

(b) Describe the purpose of checking the propeller 'track' and 'balance'.

13. (a) Describe the granodizing processes.

Or

(b) Explain various types of corrosion in detail.

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54

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aircraft Maintenance Science

EXECUTIVE COMMUNICATION

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. What are the different types of communication network?
2. Define: stereotype.
3. What is the purpose of sales promotion letter?
4. Write the significance of report writing.
5. Define: Bibliography.
6. What do you mean by Paralanguage?
7. State the principles of F2F communication.
8. What is the purpose of drafting minutes of a meeting?

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Explain about the communication barriers and how it should be overcome?

Or

(b) Describe the different phases of communication.

10. (a) Explain the structure and style of business letters.

Or

(b) Describe the various types of business letters.

11. (a) Explain in detail about the layout of a good research report.

Or

(b) Write a note on :

(i) Formal report

(ii) Informal report.

12. (a) Describe in detail about the body language and its importance.

Or

(b) Explain dyadic communication and its significance in Executive Communication in detail.

13. (a) Prepare a agenda regarding your department symposium.

Or

(b) What do you mean by drafting speech? Explain it in detail.

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61

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aircraft Maintenance Science

HELICOPTER CONFIGURATION AND MAINTENANCE

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. State the purpose of 'main rotor brake' and 'clutch' in helicopter.

2. Define the following terms :

   (a) Centre of pressure

   (b) Angle of attack

   (c) Angle of incidence.

3. What do you mean by coriolis effect and gyroscopic precession?

4. Discuss in brief about collective pitch control.

5. Write short notes on swash plate assembly fitted on helicopter.

6. Define the term "Rigging". State the purpose of tracking of main rotor blades.

7. Why do we need pitch change mechanism in tail rotor system?

8. Write short notes on "Stabilizer bar" fitted in helicopters.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Discuss in detail about Gyroscopic precession and its effects.

Or

(b) Explain the effects of torque in helicopter and state the methods adapted to counter act it.

10. (a) Explain in detail about dissymmetry of lift and state the methods of its compensation.

Or

(b) Discuss in detail about lateral movement of helicopter in air.

11. (a) Discuss in detail about tracking of rotor blades.

Or

(b) Discuss in detail about balancing of rotor blades.

12. (a) Explain in detail about rigging of tail rotor system.

Or

(b) Explain in detail about rigging of main rotor system.

13. (a) Discuss the valid points related to autorotation in general.

Or

(b) Explain in detail about "Ground effect".

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62

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aircraft Maintenance Science

COMMUNICATION AND NAVIGATION SYSTEM

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. Explain the terms wavelength and frequency in radio waves.
2. What are the various types of filters and explain?
3. What are the functions of a radio receiver?
4. What is the significance of cockpit voice recorder?
5. Write in brief about radio magnetic indicator.
6. Explain the function of market Beacon system.
7. What is modulation and explain its various types?
8. What are the advantages of MLS system?

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) Explain about various types of carrier waves in detail. (12)

Or

(b) (i) What are all the functions of a radio transmitter system? (6)

(ii) Explain the operation of superhetrodyne receiver with block diagram. (6)

10. (a) Explain the function of

(i) Localiser system. (6)

(ii) Glide slope system. (6)

Or

(b) (i) Explain the operation of aircraft ADf system. (6)

(ii) Write brief notes on SEL CAL system. (6)

11. (a) Describe about ELT, its various types, testing and maintenance. (12)

Or

(b) What is ATC transponder and explain its various modes of operation? (12)

12. (a) Describe the operation of Airborne weather radar system. (12)

Or

(b) Explain about the Radome Maintenance of radar safety. (12)

13. (a) Explain the function of GPS system with its various segments. (12)

Or

(b) Write short notes on :

(i) Aircraft DME system. (6)

(ii) Inertial navigation system. (6)

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Sub. Code
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B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aircraft Maintenance Science

AVIATION LEGISLATION – II

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 3 = 15)

Answer any **five** questions.

1. What are the conditions in which authorisations will not be issued?

2. State the procedure of evaluation of test flight report.

3. State the purpose of weighing in aircraft. Also state the occasions at which the aircraft is weighted.

4. Discuss in brief about Traffic Collision Avoidance System (TCAS).

5. Briefly discuss about minimum equipment list.

6. What do you mean by certificate of competency. Discuss in brief.

7. List out the documents related to aircraft maintenance.

8. State the requirements of airport fire station.

Part B (5 × 12 = 60)

Answer **all** questions.

9. (a) State the procedure of reporting and investigation of aircraft accidents.

Or

(b) What are the requirements of cockpit voice recorder?

10. (a) Explain in detail regarding first aid kit and physician kit.

Or

(b) Discuss in detail about the qualification and experience details of pilot carrying out test flight.

11. (a) List out the contents of flight manual.

Or

(b) Discuss in detail about operation manual.

12. (a) State the requirements for issue and renewal of flight engineers licence.

Or

(b) Discuss in detail about the requirements with regard to installation of Ground Proximity Warning System (GPWS).

13. (a) What do you mean by certificate of airworthiness, and certificate of maintenance. Discuss in detail.

Or

(b) Discuss in detail about special flight permit.

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Aircraft Maintenance Science

PRODUCTION AND MAINTENANCE MANAGEMENT

Time : 3 Hours Maximum : 75 Marks

Section A (5 × 3 = 15)

Answer any **five** questions.

1. What are the components of production management?
2. What is meant by aggregate planning?
3. Why material planning is important?
4. Define the term strategic management.
5. Why information system is needed for production planning?
6. What is the importance of reliability program?
7. What is meant by ergonomic considerations?
8. List out the importance of purchasing in production management.

Section B (5 × 12 = 60)

Answer **all** questions.

9. (a) Discuss the various types of production system with suitable examples.

Or

(b) (i) Distinguish between Standardization and Simplification.

(ii) What are the objectives of strategic management?

10. (a) Explain the process of material requirement planning.

Or

(b) Why production planning and control is important? How it increase in overall organizational efficiency?

11. (a) What is meant by TQM and Explain the fundamental factors affecting quality in production Management.

Or

(b) Describe the various techniques of quality control in production industry.

12. (a) Critically analyze the failure modes and effectiveness of information system for maintenance.

Or

(b) Explain the important steps to maintain the quality in maintenance management.

13. (a) Critically analyze the roles and components of Material and inventory management.

Or

(b) Explain the various features of inventory models.